Analyzing Austin College's Carbon Emissions and Comparing Peer Institutions

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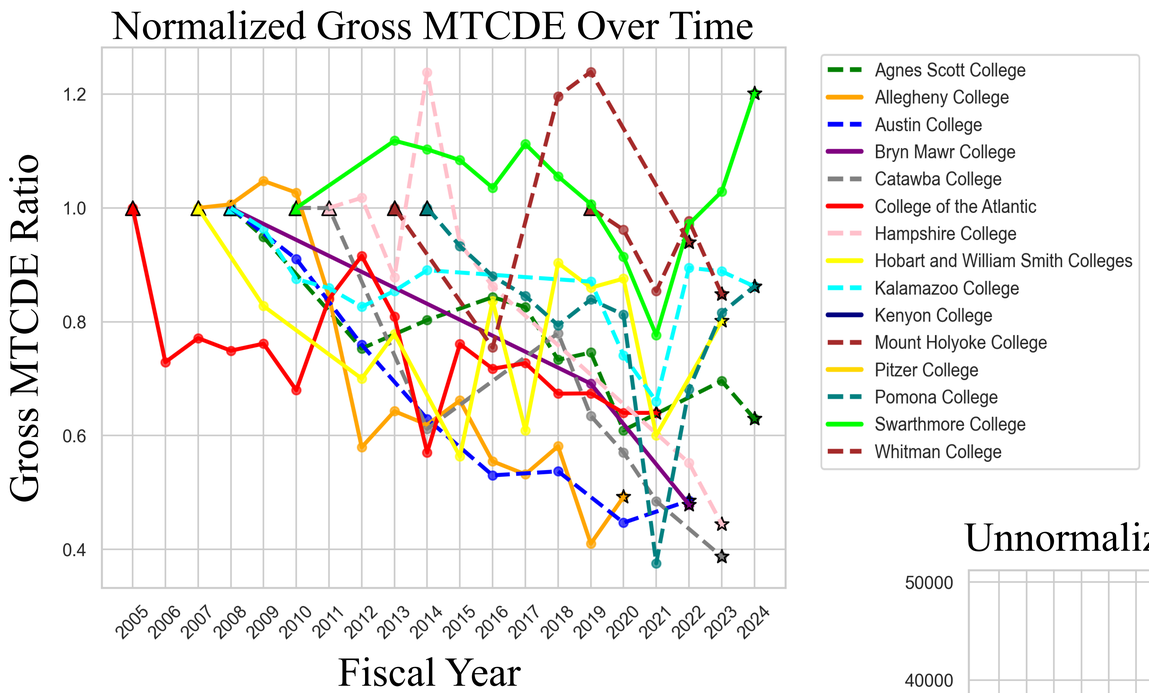
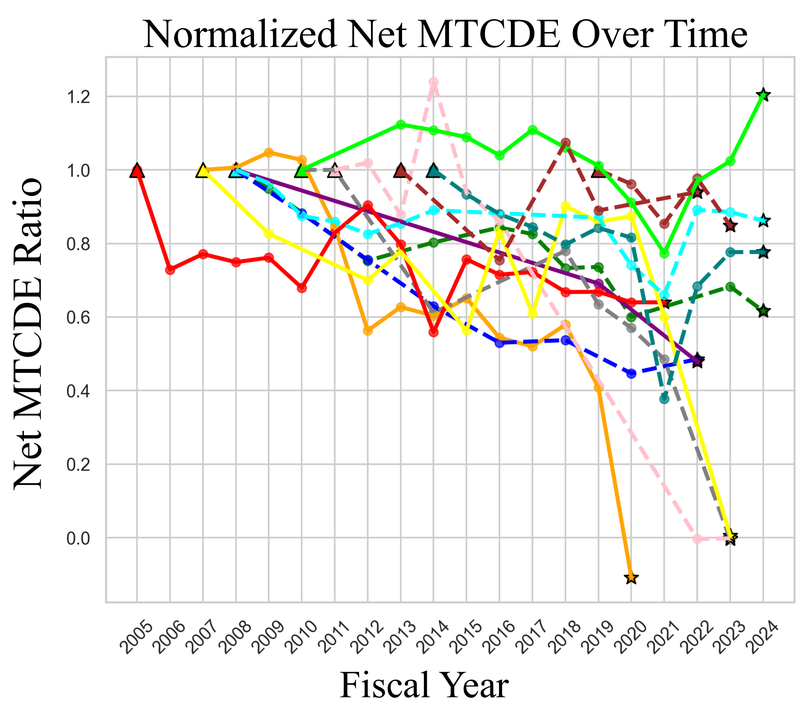
Abstract

In this work, we investigate the feasibility of Austin College (AC) achieving its goal of carbon neutrality by 2035. This project examines AC's carbon footprint using data from SIMAP (Sustainability Indicator Management and Analysis Platform) to assess progress toward that goal. The focus is on tracking AC's emissions data from 2008 to 2022 across Scopes 1, 2, and 3, as well as Gross and Net MTCDE (Metric Tons of Carbon Dioxide Equivalents). This analysis includes reductions over time and a comparison of AC's progress toward carbon neutrality with that of similar liberal arts colleges.





Three big questions

Whether AC's neutrality goal is realistic.

- Realistic with sustained funding and policy support
- Long-term commitment is essential

Which strategies (policy, offsets, green energy, student behavior) are most effective across institutions.

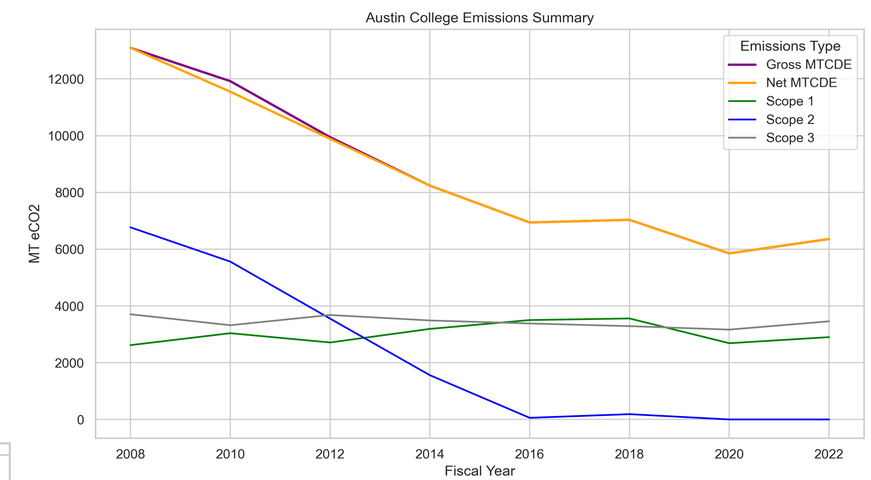
- Carbon offsets
- Reducing purchased electricity

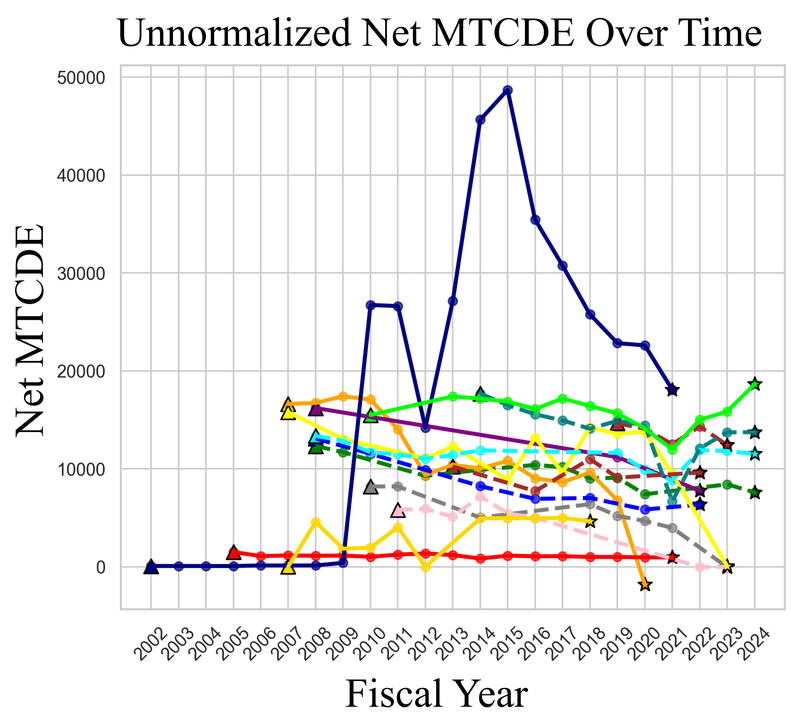
How carbon intensity (emissions per student or per building area) aligns with peers.

- Compared to similar schools, AC is reducing carbon emissions effectively.
- AC is making steady progress towards its goal.

Scopes of Emissions

- **Scope 1**: Direct emissions from college-owned sources (e.g., boilers, fleet vehicles).
- **Scope 2**: Indirect emissions from purchased electricity.
- **Scope 3**: Other indirect emissions (e.g., commuting, study abroad, air travel, waste, purchasing).





The unnormalized graph contained colleges that disrupt proper calculations for normalization. Data from Kenyon and Pitzer college had to be ignored for proper normalization. Based on individual emission reports, we saw that eighter the colleges had years with 0 emissions reported or double emissions reported. We assume that the emission reports had misconstrued data.

Conclusion

The research addresses whether Austin College's carbon neutrality goal by 2035 is achievable by:

- Documenting emission trends from 2008–2022.
- Identifying key drivers of reductions and remaining challenges.
- Comparing AC's performance with peer institutions.
- Highlighting the balance between gross and net emissions, and the role of offsets and renewable energy.

Future directions

- Develop a tutorial explaining Python techniques are used for future DSA student classes.
- Publish findings on our Ogre research website.